Question

Molecular dynamics (MD) simulations numerically solve the equations of motion for a multi-particle system described by Newton's second law:

$$m_i \frac{d^2\vec{r}_i}{dt^2} = \vec{F}_i(\{\vec{r}_j\})$$

where m_i and $\vec{r}_i$ are the mass and position of particle i, respectively, and the force $\vec{F}_i$ is typically given by the negative gradient of the potential energy function $U(\{\vec{r}_j\})$, i.e., $\vec{F}_i = -\nabla_{\vec{r}_i} U$.

Below are four numerical integration schemes for solving these equations of motion (where acceleration $\vec{a}_n = \vec{F}_n/m$).

1.

$$\begin{cases} \vec{r}_{n+1} = \vec{r}_n + \vec{v}_n \Delta t \\ \vec{v}_{n+1} = \vec{v}_n + \vec{a}_n \Delta t \end{cases}$$

2.

$$\vec{r}_{n+1} = 2\vec{r}_n - \vec{r}_{n-1} + \vec{a}_n (\Delta t)^2$$

3.

$$\begin{cases} \vec{r}_{n+1} = \vec{r}_n + \vec{v}_n \Delta t + \frac{1}{2} \vec{a}_n (\Delta t)^2 \\ \vec{v}_{n+1} = \vec{v}_n + \frac{\vec{a}_n + \vec{a}_{n+1}}{2} \Delta t \end{cases}$$

4.

$$\begin{cases} \vec{v}_{n+1/2} = \vec{v}_n + \frac{\Delta t}{2} \vec{a}_n \\ \vec{r}_{n+1/2} = \vec{r}_n + \frac{\Delta t}{2} \vec{v}_n \\ \vec{a}_{n+1/2} = \vec{a}(\vec{r}_{n+1/2}) \\ \vec{v}_{n+1} = \vec{v}_n + \Delta t \cdot \vec{a}_{n+1/2} \\ \vec{r}_{n+1} = \vec{r}_n + \Delta t \cdot \vec{v}_{n+1/2} \end{cases}$$

Under ideal numerical precision and using identical initial conditions (e.g., $\vec{r}_0, \vec{v}_0$), **which of the following schemes** would generate exactly the same **discretized** particle trajectories $\{\vec{r}_n\}$?

A. 1 and 2

B. 1 and 3

C. 1 and 4

D. 2 and 3

E. 2 and 4

F. 3 and 4

G. Except for 1

H. Except for 2

I. Except for 3

J. Except for 4

K. All of the above

L. does not exist

Answer

Correct Answer: D

Detailed Explanation

We can prove through algebraic transformation that the position Verlet algorithm (2) can be derived from the velocity Verlet algorithm (3).

CHECKPOINT 1 PTS

The position Verlet algorithm (2) can be derived from the velocity Verlet algorithm (3)

Scheme 3 (Velocity Verlet) equations:

$$\begin{cases} \vec{r}_{n+1} = \vec{r}_n + \vec{v}_n \Delta t + \frac{1}{2}\vec{a}_n (\Delta t)^2 & (3a) \\ \vec{v}_{n+1} = \vec{v}_n + \frac{\vec{a}_n + \vec{a}_{n+1}}{2}\Delta t & (3b) \end{cases}$$

Scheme 2 (Verlet) equation:

$$\vec{r}_{n+1} = 2\vec{r}_n - \vec{r}_{n-1} + \vec{a}_n (\Delta t)^2 \quad (2)$$

The most concise proof method utilizes the time-reversibility of these two algorithms. The position update formula (3a) in Scheme 3 is essentially a forward Taylor expansion. We can similarly write a backward Taylor expansion to represent the previous position $\vec{r}_{n-1}$.

1. **Write the forward position step:**

This is equation (3a) itself in Scheme 3:

$$\vec{r}_{n+1} = \vec{r}_n + \vec{v}_n \Delta t + \frac{1}{2}\vec{a}_n(\Delta t)^2 \quad \text{(Eq. A)}$$

2. **Write the backward position step:**

By replacing the time step Δt with $-\Delta t$, we can derive an expression for the previous state $\vec{r}_{n-1}$ from the current state $(\vec{r}_n, \vec{v}_n)$:

$$\vec{r}_{n-1} = \vec{r}_n + \vec{v}_n(-\Delta t) + \frac{1}{2}\vec{a}_n(-\Delta t)^2$$

Simplifying yields:

$$\vec{r}_{n-1} = \vec{r}_n - \vec{v}_n \Delta t + \frac{1}{2}\vec{a}_n(\Delta t)^2 \quad \text{(Eq. B)}$$

3. **Add the forward and backward equations:**

We sum (Eq. A) and (Eq. B):

$$\vec{r}_{n+1} + \vec{r}_{n-1} = (\vec{r}_n + \vec{v}_n \Delta t + \frac{1}{2}\vec{a}_n(\Delta t)^2) + (\vec{r}_n - \vec{v}_n \Delta t + \frac{1}{2}\vec{a}_n(\Delta t)^2)$$

4. **Simplify:**

The velocity terms $\vec{v}_n \Delta t$ and $-\vec{v}_n \Delta t$ cancel out:

$$\vec{r}_{n+1} + \vec{r}_{n-1} = 2\vec{r}_n + 2 \cdot \frac{1}{2}\vec{a}_n(\Delta t)^2$$

$$\vec{r}_{n+1} + \vec{r}_{n-1} = 2\vec{r}_n + \vec{a}_n(\Delta t)^2$$

5. **Rearrange to obtain Scheme 2:**

Rearrange the equation to isolate $\vec{r}_{n+1}$ on the left-hand side:

$$\vec{r}_{n+1} = 2\vec{r}_n - \vec{r}_{n-1} + \vec{a}_n(\Delta t)^2$$

This is precisely the expression for Scheme 2. Q.E.D.

CHECKPOINT 2 PTS

2 and 3 are equivalent

1 is a first-order method and is clearly not equivalent to the others.

CHECKPOINT 1 PTS

1 is a first-order method and is not equivalent to the others

For 4, the so-called improved Euler or midpoint method, its position update is

$$\vec{r}_{n+1} = \vec{r}_n + \vec{v}_n \Delta t + \frac{1}{2}\vec{a}_n (\Delta t)^2$$

and its velocity update is

$$v_{n+1}^{\text{midpoint}} = \vec{v}_n + \Delta t \cdot \vec{a}\left(\vec{r}_n + \frac{\Delta t}{2}\vec{v}_n\right)$$

which differs from that in 3. Thus, from the second step onward, it will produce a different trajectory.

CHECKPOINT 1 PTS

4 will produce a different trajectory from the second step onward and is not equivalent to 2 or 3